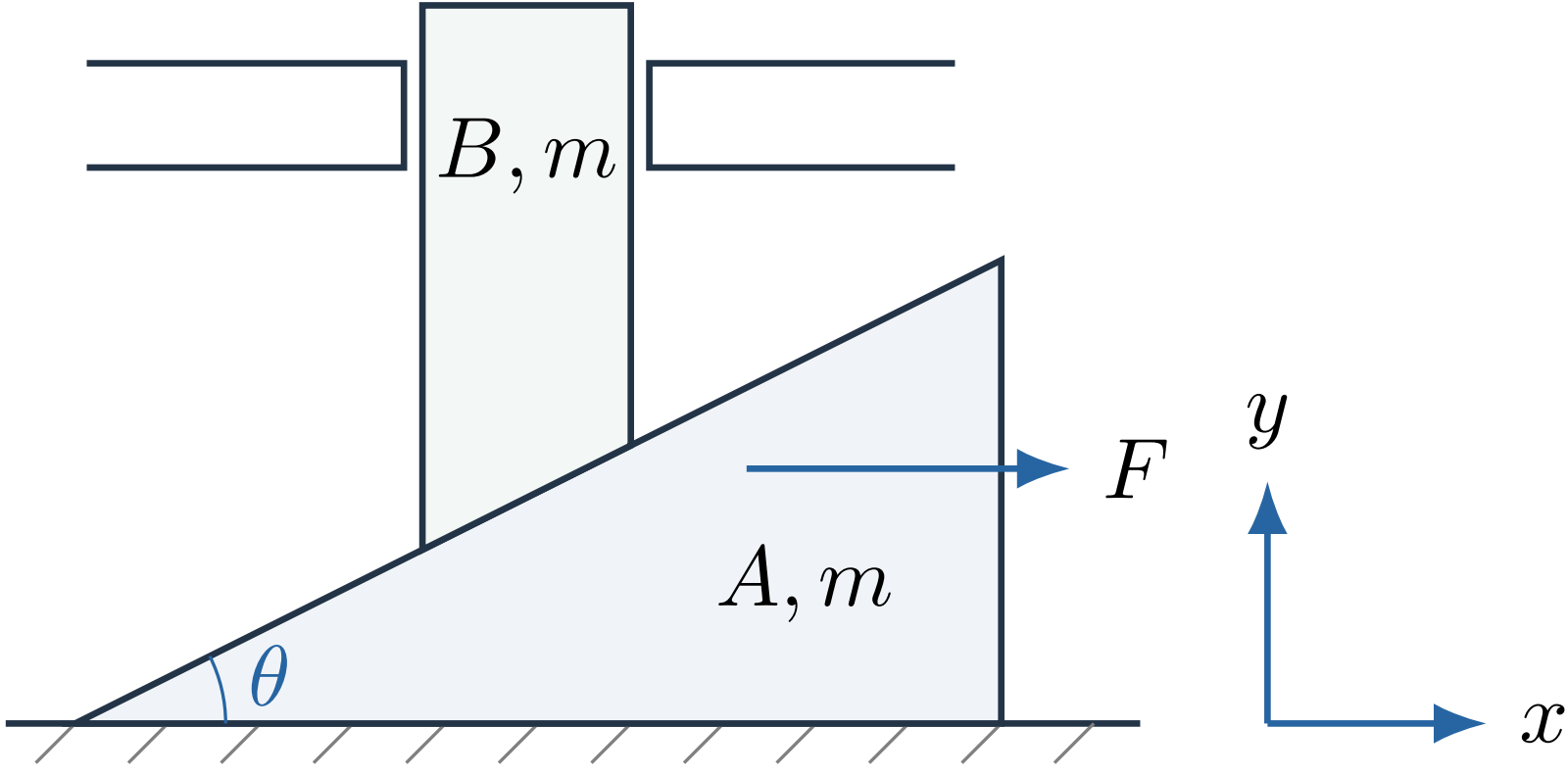
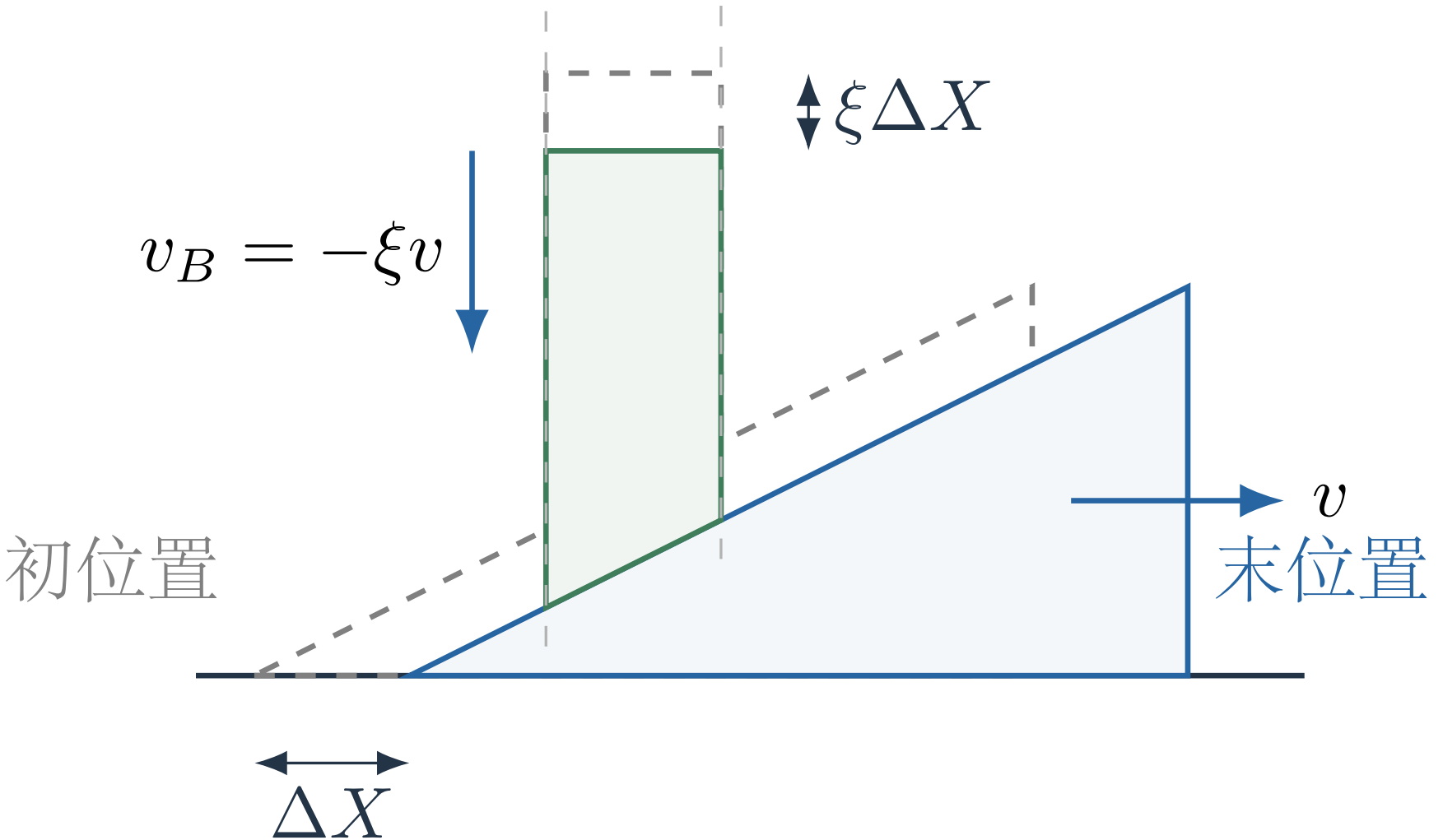
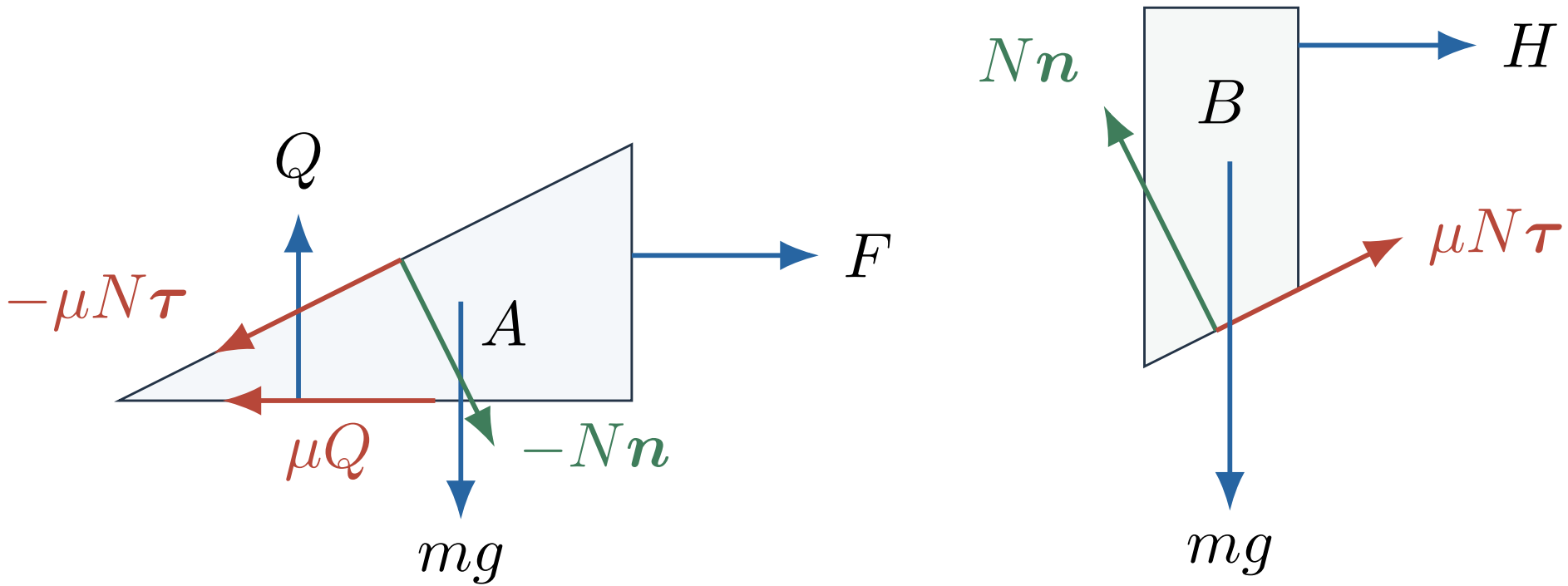
光滑竖直导向

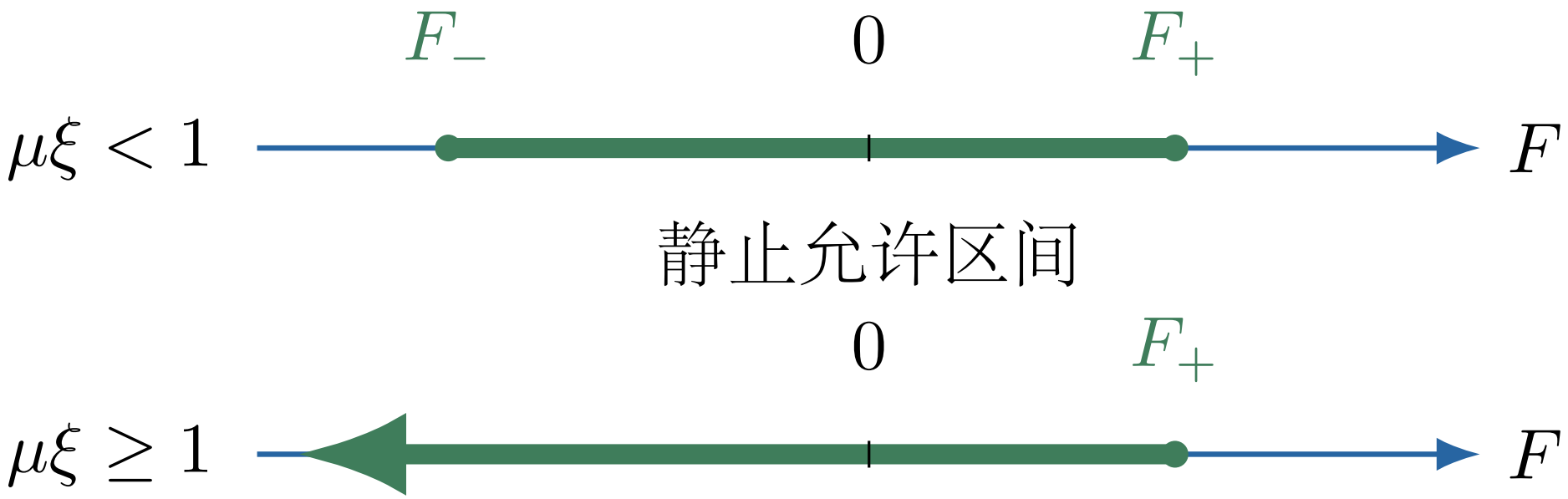


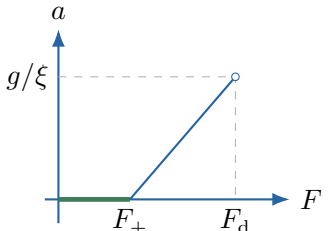


$$\xi = \tan\theta, \qquad \Delta y_B = -\xi \Delta X$$



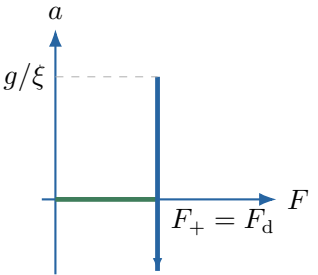
$v > 0$：接触摩擦沿相对滑动反方向


F_-
0
F_+
$\mu\xi < 1$
F
静止允许区间
0
F_+
$\mu\xi \geq 1$
F
理想自锁：负方向无有限下限



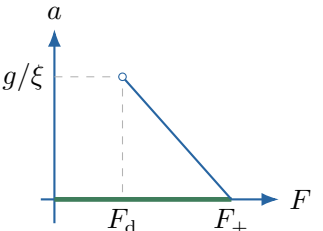
$D_+ > 0$

正常分支



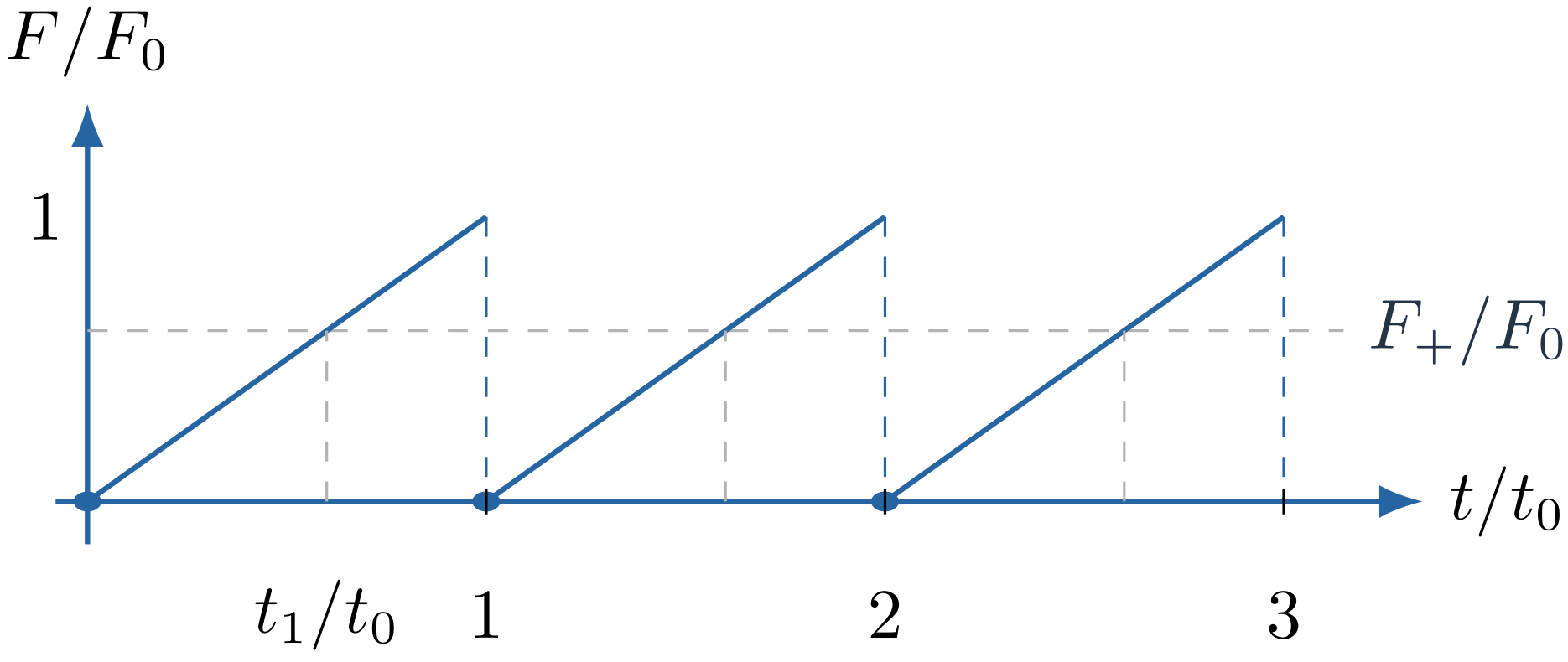
$D_+ = 0$

既有右滑动的允许集合

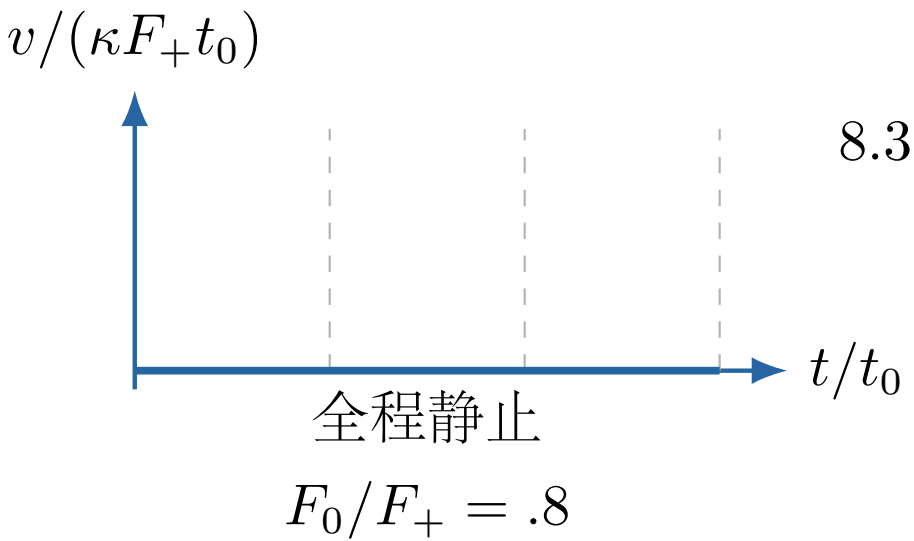


$D_+ < 0$

静止与右滑动分支并存

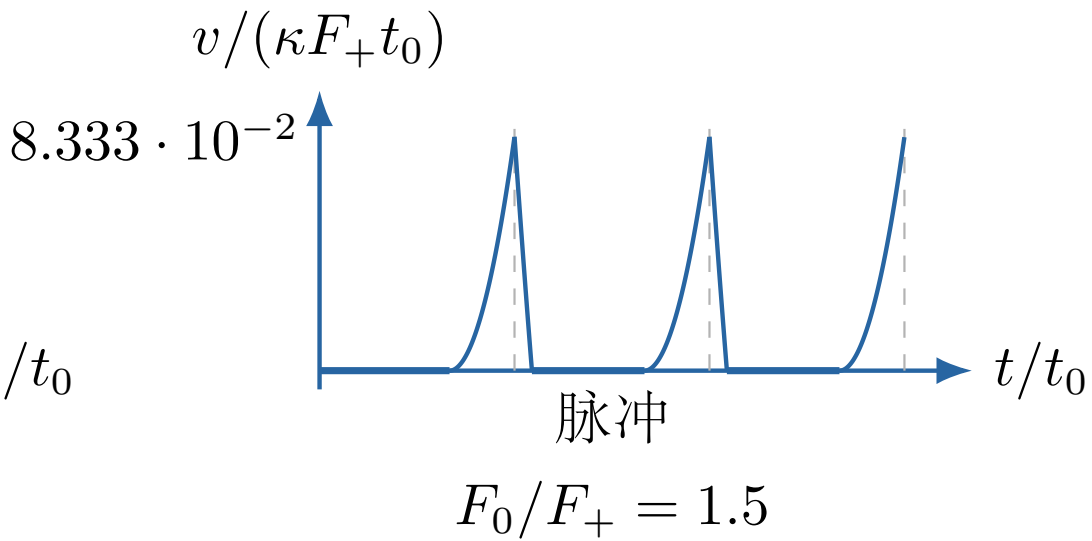


$t_1/t_0 = F_+/F_0$；力重置不造成速度跃变



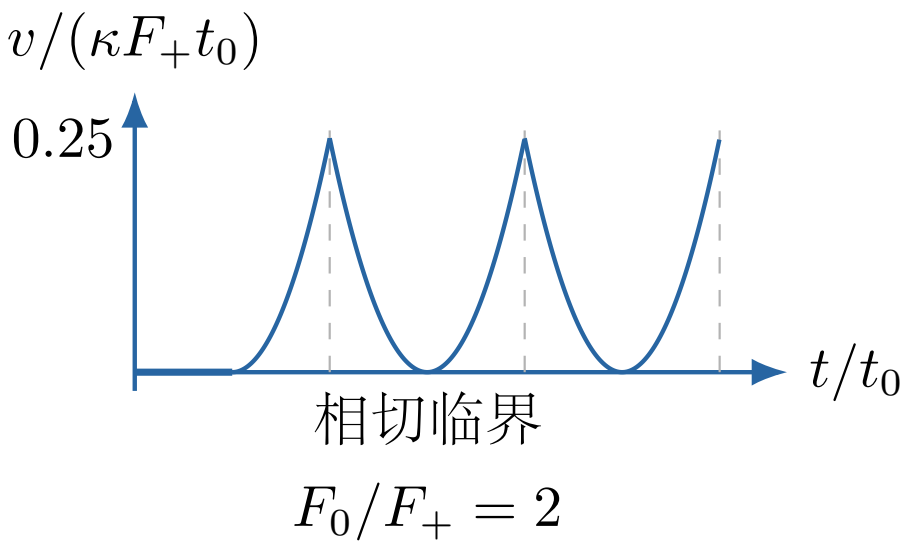
全程静止

$F_0/F_+ = .8$



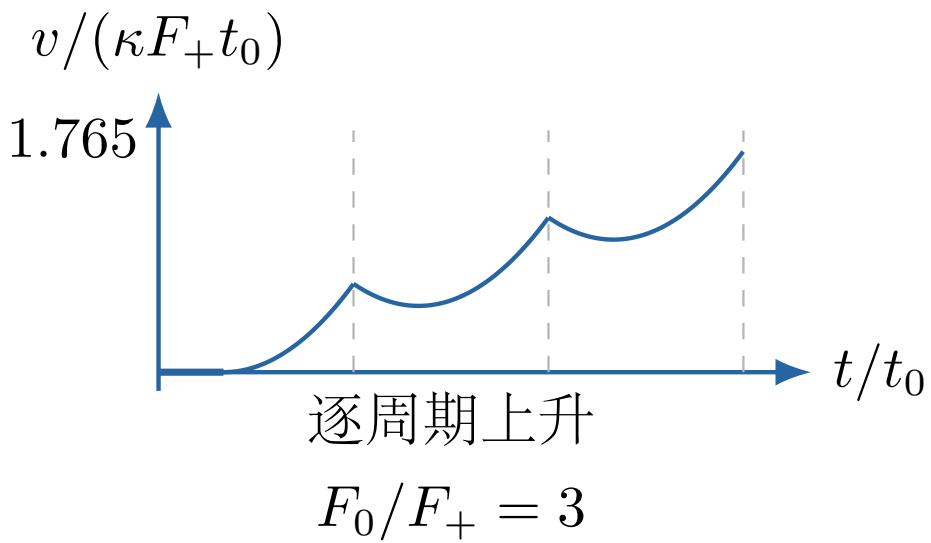
脉冲

$F_0/F_+ = 1.5$



相切临界

$F_0/F_+ = 2$



逐周期上升

$F_0/F_+ = 3$